

ELA400 Lab3

Ashley Björs* Daniel Rytenberg†

School of Innovation, Design and, Engineering

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Email: *abs21004@student.mdu.se, †drg21002@student.mdu.se,

I. THRESHOLD AND COLOR SPACE

FIND WHICH COLOR SPACE CAN ISOLATE THE IRIS THE BEST

A. Method

A good picture of the subject was taken, the quality of the image is highly dependant on the lighting as the contrast between the eyes and surrounding area needs to be high.

The approximate area surrounding the eyes was roughly cut out manually. The cropped image was put through the "Color Thresholder" app in matlab. The color spaces RGB, HSV, YCbCr & L*a*b* were tried and only HSV & YCbCr gave adequate results and was used to make two binary masks.

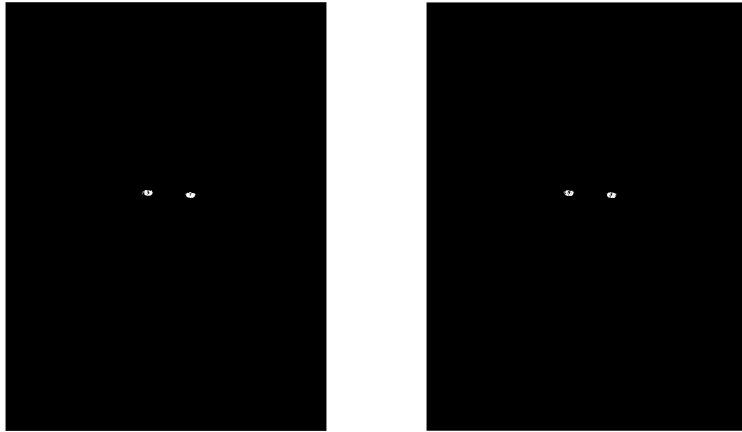


Figure 1: Left image: The Mask created using the HSV color space. Right Image: The Mask created using the YCbCr color space.

Using this mask an image with only eyes present and a image without eyes were created. The former had its RGB values shifted to make a predominantly red and a predominantly green version.

The two shifted imaged were then combined with the eyeless image to create a full image which only had a color shifted eyes.

B. Result

The resulting images of the HSV color space:



Figure 2: Left image: The Original image. Middle Image: The red shifted image. Right Image: The green shifted image.

The resulting images of the YbCbCr color space:



Figure 3: Left image: The Original image. Middle Image: The red shifted image. Right Image: The green shifted image.

C. Conclusion

The color shifting of the iris is possible but requires manual cropping, good lighting and certain color spaces to look decent.

II. CALIBRATION AND MEASUREMENT

CALIBRATE YOUR OWN CAMERA AND UNDISTORT

A. Method

23 images of a calibration image was taken from varying angles, distances and orientations and inputted into the "Camera Calibration" app. Then the outliers were removed based on the "Mean error pixels" resulting in twelve images being removed.

The app then exported a camera parameter file which was used to undistort the images. The efficacy of this undistortion was measured by comparing the length of a pen in the image and compared it to its known length.



Figure 4: The two original distorted images

The length of the pen and the length of the squares were noted

B. Result

The resulting undistorted images:

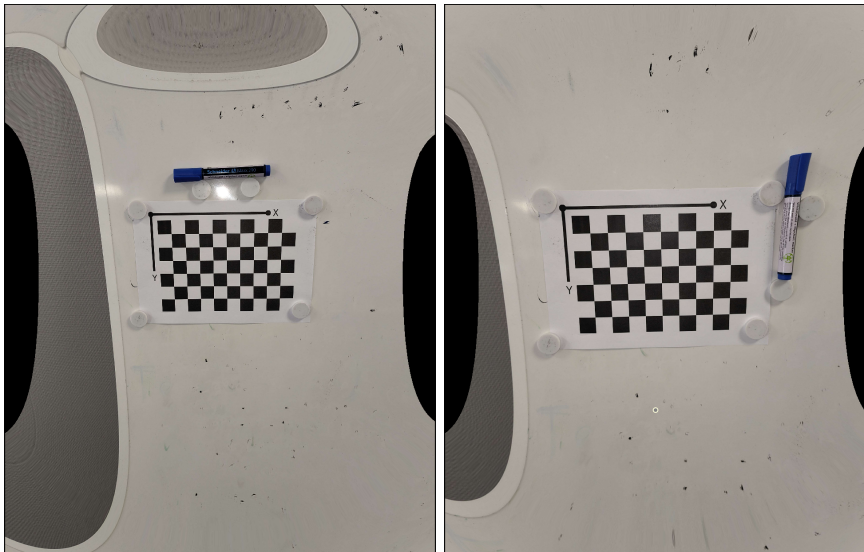


Figure 5: The two undistorted images

	Image 1	Image 2
Distorted	153.5117mm	147.7126mm
Undistorted	154.6838mm	157.2483mm

The true length of the pen was measured to be 145mm long. The undistortion resulted in a increased error from the actual pen.

C. Conclusion

Despite the tools seemingly undistorting the image the actual efficacy was negatively effected. This is theorised to firstly be due to the camera properties not being overly distorted, perhaps due to preprocessing perfomed by the camera itself. This could lead to over correction by the tool. Secondly the calibration photos could have not been up to the required quantity and quality leading to bad parameters. Thirdly the length measurement was done by hand and human error could have occured.